Assessing and Mitigating the Impact of Invasive Earthworms on Small Vegetable and Nursery Farms: An Integrated Research and Education Approach

ID: RE25-017

Link to share: https://projects.sare.org/proposals/1140376/14881/

Grant: 2025 Northeast SARE Research and Education (full proposal)

Status: Approved

Pre-proposal

Project: LNE25-489

Amount Requested: $250,375.80

Project: LNE25-489

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Description for search results if funded: This project helps small farms in VT and NY manage invasive jumping worms (JWs) through field trials and education. By using biological controls like Beauveria bassiana, cover crop, and solarization we aim to improve soil health, reduce JW impact, and enhance farm sustainability.

General Information

Project Start Date

March 1, 2025

Project End Date

February 29, 2028

Primary State

Vermont

Optional: Geographic Scope

Vermont and New York

Primary Commodities

- **None**
  - Does not apply to specific commodities

Primary Practices

- **Animal Production**
  - preventive practices
  - other
- **Education and Training**
  - demonstration
  - display
  - farmer to farmer
  - focus group
  - mentoring
  - networking
  - on-farm/ranch research
  - participatory research
  - workshop
- **Farm Business Management**
  - budgets/cost and returns

- **Natural Resources/Environment**
    - soil stabilization
- **Pest Management**
    - biological control
    - cultural control
    - field monitoring/scouting
    - integrated pest management
    - soil solarization
- **Production Systems**
    - organic agriculture
- **Soil Management**
    - earthworms
    - organic matter
    - soil analysis
    - soil quality/health
- **Sustainable Communities**
    - quality of life
    - sustainability measures

Primary Benefits and Impacts

- **Economic Sustainability**
    - Improved income or profitability
    - Increased business/enterprise opportunities
- **Environmental Sustainability**
    - Improved landscape diversity/ecological services
    - Improved soil quality/health
- **Production and Production Efficiency**
    - Improved crop production and/or production efficiency
- **Social Sustainability**
    - Improved quality of life

Was a full proposal of this work submitted previously to Northeast SARE?

- No

Does this project involve research with vertebrate animals?

- No

Does this project involve human subjects research?

- No

Plan for IACUC/IRB Determination (no word limit):

not applicable

Optional: Additional Context (150-word limit):

Small vegetable and nursery growers in the Northeast are increasingly challenged by the presence and management of jumping worms (JWs), invasive species that degrade soil structure, deplete nutrients, and disrupt plant growth. Misidentification has compounded the issue, leading some growers to waste valuable resources on pest control strategies, only to find JWs weren't present. This misallocation of time and money is particularly problematic for historically underserved and resource-limited growers who already face financial constraints and lack access to advanced pest management systems.

Additionally, fear of spreading JWs makes growers hesitant to use compost and mulch, reducing their soil health options. Our project leverages agricultural networks and extension services to provide research-based tools and education. We will offer tailored soil health assessments to help growers accurately identify and manage JWs while enhancing economic viability.

Response to Preproposal Comments (500-word limit):

We appreciate the reviewers' thoughtful input, which has strengthened our proposal.

1. **Performance Target:**

- *Change*: The performance target now focuses on specific benefits and measurable outcomes for 50-75 small-scale vegetable and nursery growers who will adopt JW management practices over 200 acres. Quantifiable benefits include improvements in soil fertility and retention of soil organic matter (SOM), translating to economic savings estimated between $29,274 and $43,910 across 200 acres.

- *Rational*: Financial impact estimates based on crop yield and SOM retention were included to address concrete metrics and verifiable benefits. We refrained from using percentages of unknown baselines, providing specific financial figures as recommended.

2. **Farming Community and Needs:**

- *Changes*: We incorporated additional evidence of farmer interest in the project's educational approach by emphasizing participatory action research and engagement during events such as NEVF and NOFA-VT. Farmers' input actively shaped the project's structure, confirming both their interest and the project's practical relevance.
- *Rationale*: We have enhanced this section to reflect how farmer interviews and participatory research shaped project methods, aligning with SARE's outcome statement for solutions addressing real-world challenges.

3. **Historically Underserved Farming Communities:**

- *Changes*: We have strengthened partnerships with women, beginning, and limited-resource farmers, and are expanding recruitment efforts with community organizations. Outreach will include targeted educational events, customized soil health assessments, and ongoing support through in-person visits and follow-ups.
- *Rationale*: This approach aligns with SARE's DEIJ values by actively supporting marginalized communities in sustainable agriculture, addressing accessibility, and promoting inclusivity.

4. **Key Individuals:**

- *Changes*: We have added farmer advisors to the project team, enhancing the project's connection to practical farming insights and supporting our educational goals. These advisors will provide feedback on management practices and host field demonstrations.
- *Rationale*: This change strengthens our team's expertise in farmer engagement and outreach.

5. **Education Plan:**

- *Changes*: We expanded the education plan to include workshops, consultations, webinars, and grower-to-grower networking opportunities, with strategies to engage farmers at multiple touchpoints. We will also offer incentives, such as SOM testing, molecular identification, and management plans, to retain and motivate farmer participation.

- *Rationale*: We focused on creating a comprehensive educational strategy that prioritizes farmer retention and addresses different learning needs and styles.

6. **Research Description:**

- *Changes*: We clarified the relevance and limitations of small-volume experimental plots by explaining our approach to scaling results for broader applicability. On-farm trials in Years 2 and 3 will validate our research outcomes in real farm conditions, bridging the gap between experimental plots and practical farming environments.
- *Rationale*: This approach aligns with the criteria by ensuring that findings are applicable across diverse farming environments while addressing scalability concerns.

Project Summary (450-word limit):

Project Focus:

This project addresses the urgent need for sustainable management of invasive jumping worms (JWs), specifically *Amynthas tokionensis*, *Amynthas agrestis*, and *Metaphire hilgendorfi*. These species are causing significant ecological and economic challenges for small-scale vegetable and nursery growers in Vermont (VT) and New York (NY). JWs degrade soil health by consuming organic matter, disrupting nutrient cycles, and leading to soil erosion, hydrophobic soils, and reduced water retention, all of which threaten crop productivity and the long-term viability of soils. The impacts are particularly severe for limited-resource growers, who face unique barriers to accessing pest management resources.

Feedback from regional gatherings like the Northeast Vegetable and Fruit Conference (NEVF) and NOFA-VT, combined with survey data by Dr. Vern Grubinger through the Vermont Vegetable and Berry Growers Association (VVBGA) and the Composting Association of Vermont, highlighted widespread concerns among growers about soil degradation, SOM loss, and ineffective pest control due to misidentification of JWs. These growers urgently need sustainable management strategies that both prevent the spread of JWs and restore soil health.

Growers have been directly involved in shaping the project, ensuring its relevance to real-world farming challenges. By addressing both ecological and economic impacts, this project aims to improve farm productivity, profitability, and soil health, especially for limited-resource growers.

Solution and Approach:

This project will implement a combined research and education program to offer growers proven, cost-effective strategies for managing JWs while improving soil health. Field trials on ten to twenty farms across VT and NY will test the effectiveness of *Beauveria bassiana* (a fungal biological control), mustard cover cropping, and soil solarization treatments.

These trials are designed to produce robust, scalable data on the success of JW mitigation techniques and their effect on SOM retention, ensuring the results are applicable across diverse farming conditions.

The educational component of the project will provide hands-on training, workshops, and field days to engage growers directly. By working with local agricultural cooperatives and extension services, the program will include tailored outreach, personalized farm visits, and educational resources to ensure participation from underserved communities. Grower-to-grower learning opportunities will foster peer-led knowledge exchange, helping growers build community resilience and long-term sustainability. Growers will also receive molecular identification for earthworm specimens that cannot be identified morphologically, along with detailed soil health assessments to monitor improvements in SOM and farm productivity over time.

Collaborating with institutions such as the University of Vermont Extension, UVM Extension Community Horticulture Program, Cornell Cooperative Extension, and New York Invasive Species Research Institute, the project aims to support farm sustainability and economic success. Ultimately, this project not only addresses immediate pest management needs but also supports Northeast SARE's goals of enhancing environmental sustainability, improving grower quality of life, and fostering community-based learning.

Performance Target

Performance Target (100-word limit)

By the project's end, 50-75 small-scale vegetable and nursery growers in VT and NY will adopt sustainable JW management techniques across 200 acres. These practices- including solarization, biological control, cover cropping, and education on JW prevention- will enhance soil fertility and increase SOM retention by 60-90%[1], leading to economic savings of $146 to $219 per acre[2]. Growers could avoid $29,274 to $43,910 in losses across 200 acres, excluding labor, energy, and input costs. Changes will be verified through biannual soil tests and financial tracking, ensuring measurable improvements in soil OM retention, fertility, and farm profitability.

Farming Community and Needs

Historically Underserved Farming Communities

- Women
- Beginning farmer or rancher
- Limited resource farmer or rancher

Farming Community (500-word limit)

The primary beneficiaries of this project are vegetable and nursery growers in New England. As part of our ongoing participatory-action research, we regularly engage with New England vegetable growers, including women, beginning, and limited-resource farmers. Annually, we meet with ~20 growers to discuss current issues, research, and co-create projects that directly address their needs. Recent meetings and surveys (Composting Association of VT Annual Meeting, Vermont Nursery & Landscape Association, VVBGA) revealed key concerns, especially from organic growers, about JW impacts, reduced SOM, soil hydrophobicity, and challenges in managing and identifying JWs. These growers, reliant on sustainable practices, face significant barriers in accessing pest management and soil health resources. This project aims to address these challenges by providing practical, cost-effective solutions for JW prevention and management, with a focus on accurate identification and mitigating risks from horticultural materials. Growers face two major challenges: some are already struggling with severe JW infestations, while others fear future invasions. For those already affected, SOM loss, erosion, and soil degradation threaten long-term productivity. For those without infestations, the primary concerns are prevention, proper identification, and avoiding unnecessary treatments. These needs were identified through regional gatherings like NOFA-VT, surveys, and direct feedback from growers. A specific case from a New Hampshire grower, who spent $900 on steam sterilization for high tunnels after mistakenly identifying JWs, illustrates the urgent need for accurate identification and cost-effective management strategies.

For growers already facing infestations, we will promote proven practices such as solarization, biological control, and compost heat treatment to mitigate soil degradation and improve SOM retention. For growers without infestations, the focus will be on preventive education to avoid future issues and conserve resources. This focus on improving soil health aligns with the growers' long-term sustainability goals, particularly for underserved growers, who often rely on SOM to maintain soil fertility without the need for expensive chemical inputs.

Growers have been involved in this project's development from the beginning. Feedback from surveys, participatory action research meetings, and direct consultations has shaped the design of the management techniques and educational outreach. For example, the misidentification of JWs and challenges in implementing organic soil management were highlighted by growers and have directly informed the project's structure. Ongoing grower input will continue to guide the project, ensuring that the solutions meet the diverse needs of the farming community.

Optional: Farmer Community Letters of Support (Upload)

AndyJones_SupportLetter

Mark&KristinKimbell_SupportLetter

GreenMountainGirlsFarm_SupportLetter

CedarCircleFarm_SupportLetter.

TreeBoard_SupportLetter

Solution and Benefits

Solution and Benefits (400-word limit)

The proposed solution focuses on integrating proven monitoring, prevention and management practices—including solarization, and biological controls—with a strong educational component to support small-scale growers in managing JW infestations. These methods are adaptable to the needs of small farms in the Northeast and aim to improve soil health, reduce crop losses, and enhance farm profitability through sustainable practices.

Benefits to Growers: The project directly addresses growers' need for practical, cost-effective strategies to manage invasive JWs. Misidentification of JWs has led to wasted resources on unnecessary treatments. This project will offer hands-on training and tools that empower growers to correctly identify JWs, monitor, understand their lifecycle, and apply sustainable management methods. Practices like solarization and biological control will help mitigate soil degradation, control JW populations, and protect farm productivity. For nursery growers, this solution will improve market confidence by minimizing the risk of JW spread through soil amendments.

How the Solution Meets Grower Needs: Growers have expressed a need for effective strategies to manage both existing JW infestations and prevent their spread through farm inputs like compost and mulch. This project addresses these needs with scalable practices suited to small farms and nurseries, promoting long-term soil health and resilience. Although environmental conditions may impact the effectiveness of certain approaches (e.g., solarization and biological controls), the project’s three-year research trials, paired with on-farm tests, will refine and validate these methods for broader application.

Contribution to Northeast SARE’s Outcome: This project aligns with Northeast SARE’s outcome statement by promoting sustainable, non-chemical management practices that improve soil health, farm resilience, and profitability. By increasing SOM retention, improving soil fertility, and ensuring nutrient availability, the project contributes to the long-term environmental and economic sustainability of Northeast farms. These practices help build farm profitability while preserving natural resources.

Research Component: The project includes a research component to validate the long-term effectiveness of prevention and management strategies such as solarization and biological control methods, such as the use of *Beauveria bassiana*. However, the success of the project does not depend on these research outcomes. The educational and outreach activities are based on proven, practical solutions that can be implemented immediately to address growers’ needs. The research will enhance future education and validate management strategies.

Project Team

Key Individuals (600-word limit)

- **Project Director: Dr. Maryam Nouri-Aiin**
  Originally from Iran, Dr. Maryam Nouri-Aiin came to the United States as a graduate student and has since established herself as a vital contributor to sustainable agriculture in the Northeast. Now a Postdoctoral Associate at the University of Vermont (UVM), Maryam is an expert in IPM, specializing in biological control strategies to manage pests such as JWs. Her interdisciplinary research uniquely integrates soil health, pest ecology, and sustainable farming practices, contributing to a stronger scientific foundation for biocontrol methods that address the unique challenges faced by the Northeast region. Her research has yielded numerous high-impact publications, establishing her as a thought leader in the field.

  As Project Director, Maryam will oversee all aspects of this project, from designing and implementing field trials to developing data-driven pest management practices and coordinating educational outreach to farmers. Her extensive experience in organizing and leading workshops on sustainable management practices makes her particularly well-equipped to foster knowledge transfer and engage diverse farming communities. Maryam ensures that this project will deliver measurable improvements in soil health, farm resilience, and economic viability for small and medium-scale vegetable/nursery farms.

- **Co-Principal Investigator: Dr. Tim McCay**
  As a Professor of Soil Ecology at Colgate University, Tim's role in the project includes supporting field trials, providing expertise in earthworm ecology, and contributing to data analysis. Known for his work in mapping and researching invasive earthworm populations, Tim will lead educational and outreach activities, particularly within New York's farming community. His experience in public education and dedication to sustainable pest management make him a valuable asset, and he is enthusiastic about contributing to improved soil health and farm resilience in the region.

- **Co-Principal Investigator: Dr. Vic Izzo**
  An entomologist and Senior Lecturer at UVM, Vic specializes in pest control strategies tailored for sustainable agriculture. With substantial experience in designing and implementing field trials, Vic will ensure that the project's research and education components are scientifically rigorous and practically applicable. He is committed to improving soil health and farm sustainability through hands-on, field-based research and looks forward to engaging growers with innovative solutions.

- **Co-Principal Investigator: Dr. Josef Görres**
  As Professor of Ecological Soil Management at UVM, Josef will play a key role in this project by assisting with field trials, contributing his extensive expertise in biological control strategies, and leading educational workshops for growers. With over 15 years of experience studying the ecology and control of JWs, Josef is also active in outreach to growers, regulators, and programs like Master Gardener and Composters. His position as Director of the Healthy Soil Collaborative gives him access to a wide network of researchers and outreach specialists. Fully committed to the project's goals, Josef looks forward to advancing soil health and resilience in the agricultural community through this collaborative effort.

- **Co-Principal Investigator: Debra (Deb) Heleba**
  As the statewide Outreach and Education Program Manager for UVM Extension's Community Horticulture Programs, Deb will focus on grower and gardener outreach, ensuring the project reaches underserved farming communities. With over 30 years at Extension and deep experience in developing and evaluating programs for commercial growers, Deb's extensive network connects home and community gardeners with the commercial landscape and nursery sectors targeted in this project. Her background as a former Northeast SARE communications specialist and her commitment to supporting underrepresented communities make her a critical member of the team, particularly in ensuring the wide and effective dissemination of research findings.

Letters of Commitment from Key Individuals (Upload)

McCay_LetterofCommitment

Heleba_LetterofCommitment

Gorres_LetterofCommitment

Izzo_LetterofCommitment

Project Advisory Committee (350-word limit)

The advisory committee for this project includes experienced growers and agricultural service providers who bring firsthand knowledge of managing JW infestations and sustainable farming practices. These individuals will provide essential input on project design, implementation, and evaluation.

Grower Members:

- **Mari Omland and Laura Olsen** (Green Mountain Girls Farm, VT): With extensive experience in diversified organic farming, Mari and Laura have dealt directly with JW infestations. They have explored management strategies such as solarization, using hens, and tillage. Their role will involve assessing grower interest, providing feedback on management practices, and hosting field days where participants can observe JW mitigation techniques firsthand.

- **Tobi Schulman** (Birds and Bee Native Plants, VT): Tobi, a beginning grower focused on biodiversity and habitat restoration through native plants, has been implementing best management practices like root washing and heat treatment to manage JWs. She will contribute insights from her heat treatment experiments and host field days for growers to see her methods in action.
- **Stacey Cooper** (Cedar Circle Farm & Education Center, VT): With over 20 years of experience in regenerative organic farming, Stacey will offer expertise on soil health and pest management. Cedar Circle Farm will host field days, showcasing sustainable JW management practices, and Stacey will provide input on the relevance and practicality of the proposed strategies.
- **Alexander Kief** (Native Plant Nursery, NY): As a new nursery owner, Alexander is particularly concerned about preventing the spread of JWs through nursery operations. He will share his experiences and challenges with JW management, contributing ideas on biological control and sustainable practices for small nurseries.

Agricultural Service Providers:

- **Dr. Deborah Neher** (Emerita Professor): A distinguished soil ecologist with expertise in biological indicators of soil health, climate change impacts, and sustainable agriculture. Her expertise in soil biology and ecology will provide invaluable insight into the project’s goals of advancing sustainable JW management practices.

The committee will meet twice annually to review progress, assess preliminary findings, and offer feedback on implementing management strategies. They will also help evaluate the project's impact on grower knowledge and adoption of best practices for JW management.

Achieving the Performance Target

Education Plan (750-word limit)

Engagement

We will leverage our networks and partnerships to engage vegetable and nursery growers in educational events, workshops, and knowledge-sharing sessions. VEPART connects with over 100 growers through direct communication and hundreds more via the VVBGA listserv, extending our reach across VT, MN, and NY. Through weekly pest monitoring newsletters, we will communicate JW prevention and management strategies.

We have secured commitments from 10 growers for participation, with ongoing recruitment through collaborations with regional cooperatives, extension networks, NOFA-VT, and NEVF. Outreach will also expand to commercial nurseries in VT and NY.

To encourage long-term engagement, we will offer incentives like complimentary SOM and nutrient levels assessments and tailored management plans. Molecular identification of JW juveniles, impossible to detect morphologically, will help growers accurately identify early-season infestations. Follow-up consultations will offer personalized support as growers implement prevention measures and best practices. A dedicated email list will facilitate ongoing communication, allowing growers to ask questions, share experiences, and collaborate on JW management. Peer-to-peer networking will be encouraged through newsletters, field demonstrations, and conferences to promote shared learning and sustainability.

Growers will receive in-person consultations, remote support, and follow-up assessments to overcome challenges. Tailored advice will be provided to growers struggling with JW prevention, such as managing compost and mulch inputs.

Learning

The educational content will focus on proven, research-backed strategies for managing and preventing JWs while improving soil health. We will emphasize practical, field-tested methods that growers can apply immediately, including: Solarization, biological controls using entomopathogenic fungi, prevention and monitoring measures

Growers will acquire the following knowledge and skills:

- **Accurate Identification and Monitoring** : Learn to identify JWs at all life stages using visual, behavioral, and environmental cues. Implement regular monitoring practices to track population levels and movement, preventing misidentification and unnecessary treatments.
- **Ecological Understanding**: Understand JW life cycles and seasonal activity patterns to better time management interventions and preventive measures.
- **Best Management and Prevention Practices (BMPs)**: Implement BMPs such as solarization, and biological controls, while applying preventive measures to stop JW introduction via inputs like compost and mulch.
- **Integration into Integrated Pest Management (IPM)**: Learn how to incorporate JW control and prevention into IPM, using cover cropping and minimal tillage to reduce infestation risks.

These skills will be delivered through a mix of field demonstrations, workshops, consultations, webinars, and online resources:

1. **Field Demonstrations**: Starting in Year 2, we will host annual field days where growers can observe JW monitoring, prevention, and management techniques like solarization and biological control, helping them apply these techniques effectively on their farms.

2. **Workshops and Consultations**: In-person and remote consultations will provide tailored advice throughout the project. Extension workshops, beginning in Year 2, will focus on JW identification, soil management practices, and sustainable prevention strategies, fostering collaboration among growers.
3. **Webinars and Online Resources**: Live webinars, along with recorded sessions and fact sheets, will cover topics such as JW identification/monitoring, BMPs, and prevention practices. These resources will be accessible via platforms like the VT and NY extension and VEPART website.
4. **Grower-to-Grower Communication**: Peer learning will be facilitated through PAR meetings, email lists, and newsletters. Growers will exchange ideas on JW prevention and management, creating a community of practice.
5. **Presentations and Publications**: Results will be shared at regional and national conferences, including NOFA conferences and Entomological Society meetings. A new fact sheet summarizing control and prevention strategies will be developed and widely disseminated.

Evaluation

The evaluation plan will assess grower learning, practice adoption, and project effectiveness through a mix of baseline assessments, real-time feedback, and follow-up evaluations.

1. **Baseline Assessment**: In Year 1, we will collect data on growers' soil management practices, JW challenges, and prevention measures through targeted surveys and interviews. This will establish a baseline for evaluating future improvements.
2. **Pre- and Post-Event Feedback**: For each learning event (workshops, field demonstrations, webinars), we will gather real-time feedback through interviews and exit surveys to track growers' progress in applying prevention techniques and management strategies.
3. **Field Visits and Follow-Ups**: Every six months, field visits and check-ins will assess how well growers are applying new practices. These evaluations will monitor changes in soil health (e.g., SOM levels), BMP adoption, and JW prevention. Personalized feedback will refine prevention efforts.
4. **Annual Reviews**: At the end of each year, surveys and interviews will assess the long-term impacts on soil health and the adoption of prevention and management practices.
5. **Post-Project Evaluation**: After the project concludes, summative evaluations will document overall success, including economic and environmental benefits like cost savings from prevention efforts and improvements in soil health.

Optional: Curriculum/Supporting Documents (Upload)

[Curriculum:Supporting Documents]

Draft Verification Tool (Upload)

[Verification Tool]

Milestones (900-word limit; suggested limit of 12 milestones, 75-word limit for each milestone)

1. **Engagement: April 30, 2025**

Educational Program Announcement and Email List Initiation

Action: Announce the educational program and initiate an email list through networks such as VEPART, VVBGA, MOFGA, NOFA-VT, UVM, Cornell Extension, and direct solicitations via peer-to-peer networks.

Participants: 100 - 200 growers

Outcomes: Growers express interest and sign up for the JW educational program.

Evaluation: Track sign-ups for the email list.

Lead: Nouri-Aiin, Heleba.

2. **Engagement: May 30, 2025**

Grower Recruitment

Action: Recruit growers through listservs and networks (VEPART, VVBGA, MOFGA, NOFA-VT, UVM, Cornell Extension), direct solicitations, and peer-to-peer networking.

Participants: 10 - 15 additional participants for field trials, supplementing 10 growers already committed.

Outcomes: Growers commit to hosting simplified field trials and agree to baseline assessments of soil health and JW infestation levels.

Evaluation: Track participant sign-ups and field trial commitments.

Lead: Nouri-Aiin, McCay, Izzo, Heleba.

3. **Engagement: June 30, 2025**

Project Launch

Action: Launch the project with a workshop.

Participants: 50 - 100 growers.

Outcomes: Growers attend workshops, gain knowledge in JW identification, monitoring, prevention, and management strategies.

Evaluation: Collect contact information through a sign-up sheet for follow-up focus groups. Use pre- and post-workshop surveys to measure knowledge gain. Workshops will be recorded and made available online.

Lead: Nouri-Aiin, McCay, Izzo, Görres, Heleba, an advisory committee member.

4. **Learning: July 1 - September 30, 2025, 2026, 2027**

Follow-up Consultations (Soil Testing and Identification)

Action: Provide follow-up consultations, including six soil test incentives (treated vs. untreated areas) and molecular identification of unidentified earthworm specimens.

Participants: 20 - 30 growers.

Outcomes: Growers implement practices tailored to their farm conditions, such as targeted solarization and biological controls.

Evaluation: Conduct pre- and post-consultation surveys to assess practice adoption and soil health improvements.

Lead: Nouri-Aiin, McCay, Görres.

5. **Learning: August 30, 2025, 2026, 2027**

Field Days

Action: Host two annual field demonstrations at the research farm and at a participating farm/nursery.

Participants: 100 - 150 growers.

Outcomes: Growers gain hands-on knowledge of JW identification, monitoring, and observe the application of prevention and management strategies, empowering them to implement these techniques on their farms.

Evaluation: Collect feedback on the relevance and practicality of the techniques.

Lead: Nouri-Aiin, McCay, Görres, an advisory committee member.

6. **Learning and Evaluation: February 30, 2026, 2027**

Workshops and Webinars

Action: Hold two annual workshops/webinars.

Participants: 200 - 300 growers (virtual or in-person).

Outcomes: Growers adopt prevention practices and integrate JW management into broader pest management systems.

Evaluation: Conduct pre- and post-workshop assessments to track knowledge gain, practice adoption, and changes in JW population density, soil health indicators, and BMP adoption rates.

Lead: Nouri-Aiin, McCay, Izzo, Heleba, Görres, an advisory committee member.

7. **Learning: February 30, 2026, 2027**

Educational Materials Distribution

Action: Distribute factsheets and research briefs on JW identification, biology, monitoring, prevention, and management through VVBGA, the PAR network, and the UVM Extension, UVM Community Horticulture Program, Cornell Cooperative Extension, and NY Invasive Species Research Institute websites.

Participants: 250 - 400 growers.

Outcomes: 60-70% of growers report improved ability to identify and manage JWs.
Evaluation: Measure knowledge improvements through feedback and surveys.
Lead: Nouri-Aiin, McCay, Heleba.

8. **Engagement and Learning: March 30, 2026, 2027**

Conference Presentations

Action: Present project findings and grower testimonials at NEVF, NOFA-VT, and the Entomological Society of America conferences.

Participants: 300 - 500 growers and agricultural service providers.

Outcomes: Growers discuss results, share experiences, and refine prevention strategies.

Evaluation: Conduct follow-up surveys and interviews to assess practice adoption among participants.

Lead: Nouri-Aiin, Izzo, Görres.

9. **Engagement and Learning: December 25, 2025, February 30, 2026, 2027**

PAR Focus Group Meetings

Action: Host annual Participatory Action Research focus group meetings to showcase successful JW monitoring, prevention, and management practices.

Participants: 40 - 50 growers.

Outcomes: Growers discuss results, share experiences, and refine prevention and management strategies based on collective learning.

Evaluation: Record participant feedback on learning outcomes and practice adjustments.

Lead: Nouri-Aiin, McCay, Izzo, Görres, an advisory committee member.

10. **Evaluation: December 30, 2027**

Final Project Evaluation

Action: Conduct final project evaluations, including feedback on economic and environmental impacts.

Participants: 50 growers submitting final evaluations.

Outcomes: Growers provide feedback on soil health improvements, JW control success, and economic benefits from BMP adoption.

Evaluation: Compare baseline and final data on soil health, JW populations, and grower costs.

Lead: Nouri-Aiin, Heleba, McCay, Izzo, Görres, and advisory committee members.

11. **Post-Project Evaluation: January 30, 2028**

Long-Term Impact Assessment

Action: Conduct post-project evaluations to assess long-term impacts on soil health and JW management success.

Participants: 40 - 50 growers providing follow-up data on continued practice use.

Outcomes: Growers report long-term JW management success and soil health improvements.

Evaluation: Use post-project surveys and interviews to evaluate sustained outcomes.

Lead: Nouri-Aiin, Heleba, McCay, Izzo, Görres.

Research

Will you be conducting research as part of your project?

- Yes

Research Hypothesis or Question (100-word limit)

How effective are minimal tillage, mustard cover crops, and biological control (*Beauveria bassiana*) in reducing JWs populations compared to untreated control plots?

Can solarization and organic matter additions (mulch and compost) achieve long-term suppression of JWs populations while improving soil health indicators, such as SOM and water retention?

What are the economic and agronomic benefits of using these treatments, specifically in terms of crop yield, soil health, and cost savings from reduced soil amendment use on small farms?

Research Description (1,200-word limit)

Farmer Input

We have identified several key growers concerns regarding JW infestations. Growers lack confidence in the effectiveness of JW's management strategies on a practical scale. Solarization, for example, have generated interest, but growers are unsure how to apply these methods to prevent JW introduction or spread on farms/nurseries. Growers have also raised concerns about the proper sanitization of imported materials like compost and mulch to ensure they are worms-free. Solarization has been mentioned as a potential solution, but growers need more guidance on applying it correctly to achieve the lethal temperatures required to eliminate JW different life stages.

Growers are also interested in exploring biological controls as a long-term management strategy. They are concerned with preventing the spread of JWs to uninfested areas on their farms or nearby forests. Grower input has shaped the treatments and research in this proposal, ensuring practical, scalable solutions for sustainable JW management that can be adjusted for varying soil types, climates, and farm conditions.

Research Farm Trials (Year1-3)

We have selected the following treatments based on grower input and the need for effective and sustainable management strategies that can be scaled for application on farms:

1. No Treatment Control:
Untreated plots serve as a baseline for comparing the effectiveness of the other treatments.

2. Minimal Tillage:
Chosen for its ability to reduce soil disturbance while limiting JW activity, aligning with growers' goals of balancing soil health and pest control
Application Method: 2-3 inches deep minimal tillage of the top soil

3. Mustard Cover Crops:
Selected for its natural biofumigation properties, test their ability to naturally deter JWs and soil health benefits, meeting growers' preference for non-chemical methods
Application Method: Sow 12 lbs/acre equivalent mustard cover crops early season

4. Beauveria bassiana (Biological Control):
A sustainable alternative to chemicals, favored by growers for long-term JW management which has shown effectiveness in previous studies to suppress JWs
Application Method: Beauveria bassiana wettable powder (applied according to label instructions) will be applied twice per season (May and September) to target JW juveniles and adults.

5. Mulch/Compost Addition:
Explores growers' interest in using organic matter to mitigate JW, while enhancing soil health
Application Method: A 2-3 inch mulch layer, combined with light compost incorporation, to explore how organic matter impacts JW populations and enhances soil health.

6. Solarization:
Chosen for its effectiveness in targeting JWs at multiple life stages, addressing growers' interest in heat treatments.
Application Method: Cover plots with clear plastic sheets at different times during the season, heating the soil to 104°F for at least 90 minutes to eliminate JWs. Temperatures at varying depths will be monitored to ensure effectiveness.

Methods

The research will be conducted using twenty-five galvanized raised beds (8 ft x 2 ft x 1.4 ft) to isolate treatments and provide controlled conditions, each filled with worm-free soil. The galvanized material in these beds was selected for its ability to isolate treatments without significantly increasing soil temperatures, except for minimal edge effects. To ensure reliable results, the experiment will be conducted in a shaded area on the edge of a JW-infested forest at the UVM Horticulture Research Center. All raised beds, except those with cover crops, will be planted with lettuce. Lettuce was chosen because it is a shallow-rooted crop and susceptible to drought, allowing us to observe potential JW impacts on crop health and growth. Each bed will receive 100 JW juveniles at the start of the trial. The first five treatments (including minimal tillage, mustard cover crops, Beauveria bassiana, and mulch/compost addition) will be applied directly to these raised beds.
For solarization, we will assess its ability to achieve lethal soil temperatures (104°F for at least 90 minutes) by experimenting with insulation layers beneath solarized piles. We will test different timing windows in early spring and mid-summer to reflect real-world conditions, targeting periods when JW populations are most vulnerable in early spring and reach their highest abundance in mid-summer. The experiment will use a randomized complete block design with five replicates per treatment and control. To prevent worms from migrating between plots, aluminum mesh insulation layers will be placed beneath each bed.

Data Collection:

1. JWs Population Density: JW cocoons, juveniles, and adults will be counted every 10 days. Data will be collected from each treatment and control bed to track population changes over time.

2. Crop Performance: For plots with crops, we will monitor crop growth and yield every 10 days. Data will include germination rates, plant height, leaf size, and total crop yield at harvest.

3. Soil Health Indicators: Six soil samples will be taken at three key points: pre-treatment, mid-season, and post-harvest, to assess SOM, nutrient availability, and water retention.

4. Economic Impact Baseline: Data will be analyzed to evaluate the effects of JW on nutrient loss, input needs, crop yield, water use, and treatment costs, establishing an economic baseline.

On-Farm Trials (Years 2 and 3)

In the second and third years, the two best-performing treatments from the UVM trials will be tested on 10-20 vegetable and nursery farms across VT and NY. This phase aims to validate the treatments under real-world conditions, ensuring their scalability across farms of different sizes. Growers will apply one of the selected treatments to 3 treated and 3 untreated 10 ft x 10 ft plots, separated by weed mat barriers to prevent cross-infestation.
Baseline soil data (pH, texture, organic matter) will be collected to ensure consistency. This data will also be used as covariates in the final analysis to control for environmental differences across sites. Growers will be responsible for collecting data on worm populations, soil samples, and crop productivity.

Data collection will focus on:

1. JWs Population Density: Growers will count JWs in treated and control plots at early, mid, and late season intervals.
2. Soil Health Indicators: Growers will collect three soil samples from treated areas and three from untreated areas, and we will assess SOM, nutrient availability, and water retention using simple, grower-friendly methods.
3. Crop Yields: Growers will measure crop yield and productivity, comparing treated and control plots to assess the impact of the treatments.
4. Cost Comparison of Soil Amendments: Growers will track soil amendment usage and costs in treated vs. untreated plots. Data will be analyzed to assess the effects of JW on nutrient loss, input needs, crop yield, water use, and treatment costs, establishing a baseline for economic impact.

Data Analysis
Both research and on farm trail data will be analyzed using linear mixed models in RStudio, with treatments as fixed effects and blocks as random effects. If data are not normally distributed, generalized linear mixed models will be applied. For the on-farm trials, farms will be added as a fixed effect to account for environmental variability, providing insights into how treatments perform under real-world conditions.

Additional Information
The research plots will serve as demonstration sites during field days, where growers can observe JW management techniques. On-farm trial participants will share their experiences, with the process documented through photos, videos, and shared via online platforms and grower networks. We will share findings—through field days, farmer networks, technical reports, and online resources—ensuring farmers across diverse regions can apply the results to their specific contexts.

Optional: Other Relevant Research Information (Upload)

Other Relevant Research Information

Previous Work

Previous Work (750-word limit)

The management of invasive earthworms (*Amynthas* species) is a growing concern for small farms and nurseries in the Northeast U.S. Nurseries are especially vulnerable due to their reliance on imported horticultural materials, which can introduce JW. JWs significantly degrade soil health by consuming organic matter, disrupting nutrient cycling, and negatively altering soil structure and water retention [3–10]. Invasive earthworms can reduce SOM by up to 90% [1], leading to significant nutrient loss. For instance, each 1% of SOM provides approximately $48.79 in value from nutrients like nitrogen, phosphorus, potassium, and sulfur [2]. A 90% reduction in SOM can result in economic losses as high as $219.55 per acre annually. While research on JW management in agricultural systems is still evolving, insights from related studies on invasive earthworms and soil health offer a valuable foundation for this project.

Current Knowledge and Practices:

Studies have demonstrated that JWs disrupt ecosystems by rapidly consuming organic matter and altering soil composition [1,11]. Their activity increases soil erosion, reduces SOM, and leads to hydrophobic soils, making it difficult for water to penetrate and be retained [9,10,12–15]. The decline in SOM also affects nutrient cycling, making soils less fertile and less able to support healthy crop growth. In forest ecosystems, JWs have been linked to reduced biodiversity and hindered forest regeneration [5,8,11,14,16–20], but their impact on agricultural systems—particularly small farms—remains understudied.

Farmers and nursery operators have expressed the need for effective management strategies, particularly in preventing the spread of JWs and mitigating their effects once present. Horticultural materials including plant soil, mulch and compost are the main sources of spreading JWS [21]. Current methods include practices like solarization, which involves covering the soil with plastic to trap heat, raising the temperature to levels lethal to worm cocoons, the cryptic side of their life cycle [22]. Our preliminary data (unpublished, Nouri-Aiin et al.) shows that heat treatment at 104°F for 90 minutes effectively kills 100% of both adult worms and cocoons. This promising result underscores the potential of thermal control methods for managing JWs [23]. The next phase of our research will focus on adapting this treatment to soil solarization, testing its efficacy on a larger scale in diverse farming environments. *Beauveria bassiana* as a biological control agent, has being explored for its potential to manage worm populations [24,25]. However, these strategies have yet to be widely adopted or fully tested in small-scale farming environments.

Research and SARE-Funded Projects:

Several SARE projects have explored vermicomposting, but only SARE project

YENC09-013, *The Early Bird Gets the Worm* [26], specifically addressed the role of worms in soil fertility while highlighting the risks of invasive species in agricultural and ecological contexts. Initially, worms were used to speed up the decomposition of organic matter, but the project found that invasive species could cause nutrient leaching and harm soil health. This project emphasizes the need to recognize and manage the ecological risks posed by invasive species like JWs, offering important insights into soil fertility and organic matter management.

Existing Efforts:

Our project builds on these prior research efforts by directly applying known management techniques—solarization, minimal tillage, cover cropping, and biological controls—specifically to the challenge of managing JWs in agricultural systems. While solarization has been successfully tested in controlled environments, our project will adapt this method for practical use on small farms. By incorporating biological controls such as *Beauveria bassiana* into existing farming practices, we aim to offer a comprehensive, sustainable solution that addresses the specific needs of vegetable and nursery growers in VT and NY.

The project is also informed by grower feedback gathered through surveys and focus groups. Farmers have expressed uncertainty about how to effectively implement solarization and other management practices, and they have voiced concerns about the potential spread of JWs through soil amendments such as compost and mulch. Our project addresses these gaps by providing hands-on demonstrations and educational resources that are tailored to the scale and resources available to small farms.

Adaptation:

We emphasize the adaptation of JW management strategies for vegetable and nursery operations in the Northeast. We will test these methods in real-world farming conditions to ensure they can be effectively applied without disrupting other essential farm activities. Additionally, we will integrate JW management into broader sustainable farming practices, such as IPM and soil health management, ensuring that growers can adopt these practices without compromising the productivity or ecological health of their farms. This holistic approach will help protect soil quality while providing long-term solutions for controlling JWs.

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Budget

Organizational Affiliation of Partners

Timothy (Tim) McCay

Colgate University (Other college or university)

Budget Justification and Narrative Spreadsheet (Upload)

LNE25-489 Nouri-Aiin, University of Vermont Budget FINAL

Total Direct Costs

227,614.36

Total Indirect Costs

22,761.44

Total SARE Request

250,375.80

Host Organization Approval

Grant Commitment Form (Upload)

Nouri_NortheastSARELargeGrantsCommitmentFormVersion_signed

ColgateNortheastSARELargeGrantsCommitmentFormVersion

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FDP Clearinghouse Information

- Yes

US Department of Agriculture

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NORTHEAST SARE Sustainable Agriculture Research & Education

Grant Commitment Form

For the following Northeast SARE grant programs:

Professional Development Program

Research and Education

Research for Novel Approaches

This form must be completed with signatures and attached to the online application at time of submission. Applications will not be accepted without the fully signed Commitment Form, nor will Commitment Forms be accepted after the submission deadline of ***5:00 p.m. ET on October 29, 2024.***

Project title: Assessing and Mitigating the Impact of Invasive Earthworms on Small Vegetable and Nursery Farms: An Integrated Research and Education Approach

Total funds requested that would go to the organization/institution/business: $ 13,764

Assurance of Project Leader/Collaborator

For this project, I affirm that I am, or will be, an employee or authorized representative of:

Colgate University *(organization/institution/business to receive the proposed funding).* Should this proposal be awarded, I will be the primary contact at my organization/institution/business for managing the project/subaward. I will be responsible for reporting project results by January 15 each year while the project is in progress and providing a detailed final report within 60 days of the project's end date. I will acknowledge Northeast SARE as a funding source in all project publications and outreach materials. I will keep Northeast SARE informed of any contact and e-mail changes for at least two years after the final report is submitted.

Does this project involve human subjects research?

☐ Yes ■ No

Does this project involve research with vertebrate animals?

☐ Yes ■ No

If I checked yes to either of the above, I understand that I will be required to obtain an IRB or IACUC determination and submit evidence of that process to Northeast SARE, prior to any funds being provided for research.

Signature of project leader or collaborator: DocuSigned by: Timothy S. McCay 3BE603DE47B84E0... Date: 10/15/2024

Print name of project leader or collaborator: Timothy McCay

Please proceed to next page for Organizational approval.

Organizational approval:

As the authorized grants or sponsored programs official, or the chief financial officer/owner (for a business), of Colgate University _(name of organization/institution/business)_ I hereby certify that I have reviewed this proposal and approve the funding request as defined in the Budget Justification and Narrative, and confirm that we have the capacity to manage grant funds on behalf of the project leader or collaborator named above, should the proposal be funded.

I further understand that any the Northeast SARE funds awarded for this project to our organization/institution/business cannot be used except as outlined in the proposal.

Additionally, I am aware of whether this project involves either human subjects research or research with vertebrate animals.

Signature of authorized official: DocuSigned by: Brittany Plumley 54D3CC3CE81040F... Date: 10/17/2024

Printed name and title of authorized official: Brittany Plumley

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Is this institution registered in the Federal Demonstration Partnership (FDP) Expanded Clearinghouse?

☐ Yes, our organization profile can be found at: https://fdpclearinghouse.org/organizations.

■ No